# Thomas Y.L. Lin

Seattle, WA • thermaus@uw.edu • linkedin.com/in/thomaslin0908 • Website • Github • Google Scholar

## EDUCATION

**University of Washington** — Seattle, WA, USA
*B.S. in Physics, Minor in Applied Math; GPA: 3.95* — 2025 – 2027 (expected)

**National Taiwan University** — Taipei, Taiwan
*B.S. in Physics, B.A. in Economics* — 2023 – 2025 (transferred)
Honors: Dean's List Award (Physics Dept., 1st Rank, Fall 23 & Spring 25); BaBar-SURF Award (one of 8 awardees)

## EXPERIENCE

**Swallowtail Labs** — San Francisco, CA, USA
*Co-Founder* — May 2026 – Present
- Building a multi-physics (electrical-thermal-optical) world model on chips for planning their own thermodynamics.
- Partnered with MediaTek after a 90-min pitch to the Chairman, securing angel funding on the research prototype.

**California Institute of Technology** — Pasadena, CA, USA
*Summer Undergraduate Research Fellowships (SURF)* — March 2025 – January 2026
- Developed Decoupled Diffusion Inverse Solver (DDIS), a physics-aware, data-efficient generative framework for PDE inverse problems, decoupling coefficient prior learning from physics enforcement.
- Proved a guidance-attenuation failure mode in prior joint-embedding models under limited data, justifying our design.
- Achieved SOTA with 1% paired data, improving reconstruction error by 11% (L2) and 54% (spectral) over prior best solvers on inverse Helmholtz, Poisson, and Navier–Stokes problems.
- Advised by Prof. Anima Anandkumar and PhD student Jiachen Yao.

**Northwestern University** — Remote
*Machine Learning Intern* — April 2023 – April 2025
- Analyzed the fine-grained complexity of modern Hopfield models; designed approximation algorithms reducing time complexity from subquadratic to near-linear (ICML 2024).
- Built a latent Bayesian network for portfolio optimization — a data-driven Black-Litterman model learning investor views as latent variables (ICML 2025).
- Advised by Prof. Han Liu and PhD candidate Jerry Yao-Chieh Hu.

**Gamma Paradigm Capital** — Taipei, Taiwan
*Quantitative Analyst Intern* — June 2023 – July 2024
- Implemented Distributionally Robust Optimization portfolio strategies (Blanchet, Chen, and Zhou, 2022) across 11 sectors (2006–2023): Sharpe [0.68, 0.78], MDD [-18.4%, -27.1%] vs. SPY (0.55, -55.2%) and EQW (0.54, -37.2%).
- Built a data-driven Black-Litterman model: +49.8% Sharpe over Markowitz (0.66–0.87 vs. 0.35–0.62) and S&P 500 / DJIA on 20- and 30-year data; 55.1% lower turnover, robust to hyperparams. Spun out into research with Northwestern.

## PUBLICATIONS (**Equal Contribution*)

**On JEPA Isotropy** — **NeurIPS 2026**
**Thomas Y.L. Lin**, Han Liu, Jerry Yao-Chieh Hu. — Manuscript Link

**Decoupled Diffusion Sampling for Inverse Problems on Function Spaces** — **AI&PDE @ ICLR 2026 (Oral)**
**Thomas Y.L. Lin***, Jiachen Yao*, Lufang Chiang, Julius Berner, Anima Anandkumar. — ArXiv Link

**Latent Variable Estimation in Bayesian Black-Litterman Models** — **ICML 2025**
**Thomas Y.L. Lin**, Jerry Yao-Chieh Hu, Paul W. Chiou, Peter Lin. — ArXiv Link

**On Computational Limits of Modern Hopfield Models: A Fine-Grained Complexity Analysis** — **ICML 2024**
Jerry Yao-Chieh Hu*, **Thomas Lin***, Zhao Song, Han Liu. — ArXiv Link

## ACHIEVEMENTS & LEADERSHIP

**The Residency** — *Resident (f26 cohort)* — San Francisco, USA | Fall 2026
- Selected for The Residency's founder cohort (20 of 2,000 applicants).

**Conference Reviewer** — 2026
- NeurIPS 2026, ICLR 2026 (AI & PDE Workshop), ICML 2026 (AI4Science Workshop).

**Harvard Undergraduate in Asia Program (HUAP) NTU Branch** – *PR Member* — Taipei, Taiwan | 2024–25
- Fundraised a record $4,500 (4 deals) — 97% of the 7-member PR team's total.
- Hosted Harvard undergraduate students in March 2025 for HUAP's Spring Conference at NTU.

**European Physics Olympiad (EuPhO)** – *Bronze Medalist* — Ljubljana, Slovenia | May 2022
- Fundraised $14,000 for Taiwan's olympiad participation from companies, foundations, and government.

## SKILLS & INTERESTS

**Research:** Neural Operators, Generative Models, Inverse Problems, AI for Science, World Models, Control Theory
**Technical:** Python (PyTorch, Pandas, NumPy, scikit-learn, Statsmodels, Matplotlib, Hiplot), CUDA, Gurobi, MATLAB